9] Parity bit ⇒ A bit appended to an array of binary digits to make the sum of the digits including the parity bit always odd or even.

10] Intrepretation of the syndrome of Hamming code:

→ Syndrome created by the XOR of the code in a word with calculated version of the code.

Each bit syndrome is 0 or 1 according to if there is or is not a match position for the 2 input. If all 0's in syndrome → no error.

if 1 & only 1 bit set to 1, then error has occurred in one of the 4 check bits. ↓ no correction needed.

more than 1 bit set to 1 → numerical val of the syndrome indicates → position of the data bit in error.

↳ Data inverted for correction.

11) SDRAM vs DRAM

↓ ↳ asynchronous

exchange data with processor

↳ synchronized to an (external clock) signal running at full speed of the processor/mem bus.

↳ no wait states.

12) DDR RAM:

dramatically increase data rate over DRAM.

3 ways to achieve higher data rate:

1. Data transfer synchronized to both the rising & falling edge of the clock. → (double) data rate
2. Higher clock rate → transfer rate ↑
3. Uses buffering scheme.

(13) NAND vs NOR Flash Mem:

NOR: Basic unit of access → bit (called memory cell)

- cells connected parallel to the bit lines → so each cell can can be read/write/erased individually.

Similar function to NOR logic gate.

↳ if any mem cell of device turned ON by the corresponding word line
↳ bit line goes low.

NAND: organized in transistor arrays with 16 or 32 transistors in series.

NAND Logic Gate funct

↳ All transistors in the corresponding word lines turned ON.
→ bit line low

(14) 3 newer nonvolatile solid state Mem tech:

STT-RAM: Spin-transfer torque RAM →
↳ Magnetic
↳ new transfer mechanism
Magnetization of free layer → flipped by electrical current directly.

PCRAM: Phase change
↳ chalcogenide allay material
(similar → Optical storage media
compact disk, digital versatile disk)

↳ Storage capability from resistance difference between amorphous (high resistance) & a crystalline (low resistance) phase of chalcogenide material.

ReRAM: creates resistance rather than directly storing charge.

PROBLEM

(1) Why RAM 1 bit/chip & ROM multiple

=> Advantages

fewer pins on package (only 1 data out line)
↳ Higher density of bits for given size package.

1 output Driver → More reliable.

ROMS are much smaller → entire ROM in 1/2 bits.
↳ saves cost

(2) DRAM → Refresh cycle → 64 times/ms

Each Refresh operation → 150 ns

Memory cycle → 250 ns

Total refresh op in 1 ms = $64 \times 150 \times 10^{-9}$

$= 9.6 \times 10^{-6}$ s = 9.6 μs

$$\text{Total Memory cycles} = \frac{1\text{ms}}{\text{Mem cyc time}}$$

$$= \frac{1 \times 10^{-3}\ \text{s}}{250 \times 10^{-9}\ \text{s}}$$

$$= 4000.$$

Percentage of Refresh time $= \left(\frac{\text{Tot ref time}}{\text{Tot mem time}}\right)$

$= \frac{9.6 \times 10^{-6}s}{1 \times 10^{-3}} \times 100\%$

$= 0.0096 \times 100\%$

$\simeq 1\%$

(3) Read Operation of DRAM

access time $\rightarrow t_1$ to t_2

recharge time $\rightarrow t_2$ to t_3

(a) access time = 60 ns

recharge " = 40 ns

Mem cycle time = Acc t + rec t

$= 60 + 40 = 100$ ns

Maximum Data rate $= \frac{1 \text{ bit}}{100 \text{ ns}} = 10$ Mbps.

(b) 32 bit $\times$ 10 Mbps $= 320$ Mbps $= \frac{320}{8}$ MB/s

$= 40$ MB/s

4

6 Memory → 64K x 1 DRAMS

cell array of DRAM → 256 rows

each refresh Once every 4 ms.

(a) time period between successive ref req:

$$\Rightarrow \frac{4000\,\mu s}{256} = 15.625\,\mu s$$

(b) length of refresh address counter:

$\Rightarrow$ 256 row $= 2^8 \rightarrow$ 8 bit counter.

to represent all rows.

7



9 failure rate per billion device hour = FIT

MTB = Mean time between Failures.

1 MB Memory of 16bit microprocessor with 256K * 1 DRAMS.

↳ 256 Kb in each DRAM

8 Mega bits

$$\therefore \text{DRAMs Needed} = \frac{8\,Mb}{256\,Kb} = 32 \text{ DRAM CHIPS}$$

Combined Failure rate = 32×2000 FITS

$= 64000$ FITS

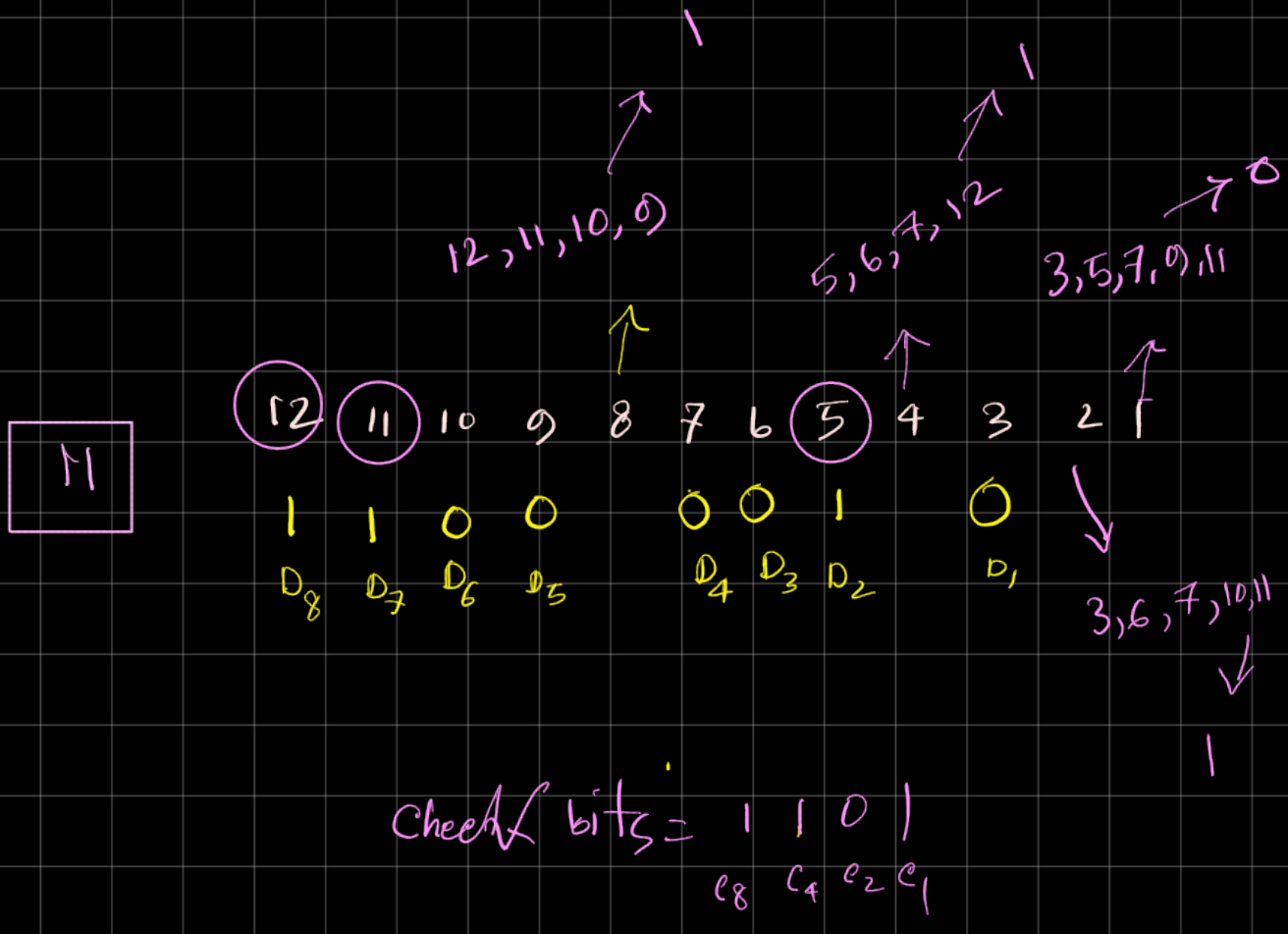
$$MTBF = \frac{10^9 \text{ hrs}}{64000} = 15625 \text{ hrs}$$

(730 hrs per months) $\therefore \frac{15625}{730} \simeq 22$ month.

10 5.10 Fig

Non zero bits → 10, 0, -7, ⑥ 3,

$$\begin{array}{r} 0111 \\ 1010 \\ 1001 \\ 0111 \\ 0110 \\ 0011 \\ \hline 0110 \end{array}$$



11

Check bits = 1 1 0 1

C_8 C_4 C_2 C_1

12

12 11 10 9 7 6 5 3

0 0 1 1 1 0 0 1

0111

1101

1101

0011

0111

Paboya!

1010

1001
1010
1010

000 1100 1

13 $2^3 = 8$ yoohoo!

$\Rightarrow n+1$

$2^{10} \Rightarrow 11$

14 16 bit $\rightarrow 2^4 \rightarrow$ Check bit = 5

D16 D11 D1

21 20 19 18 17 (16) 15 14 13 12 11 10 9 (8) 7 6 5 (4) 3 (2) (1)